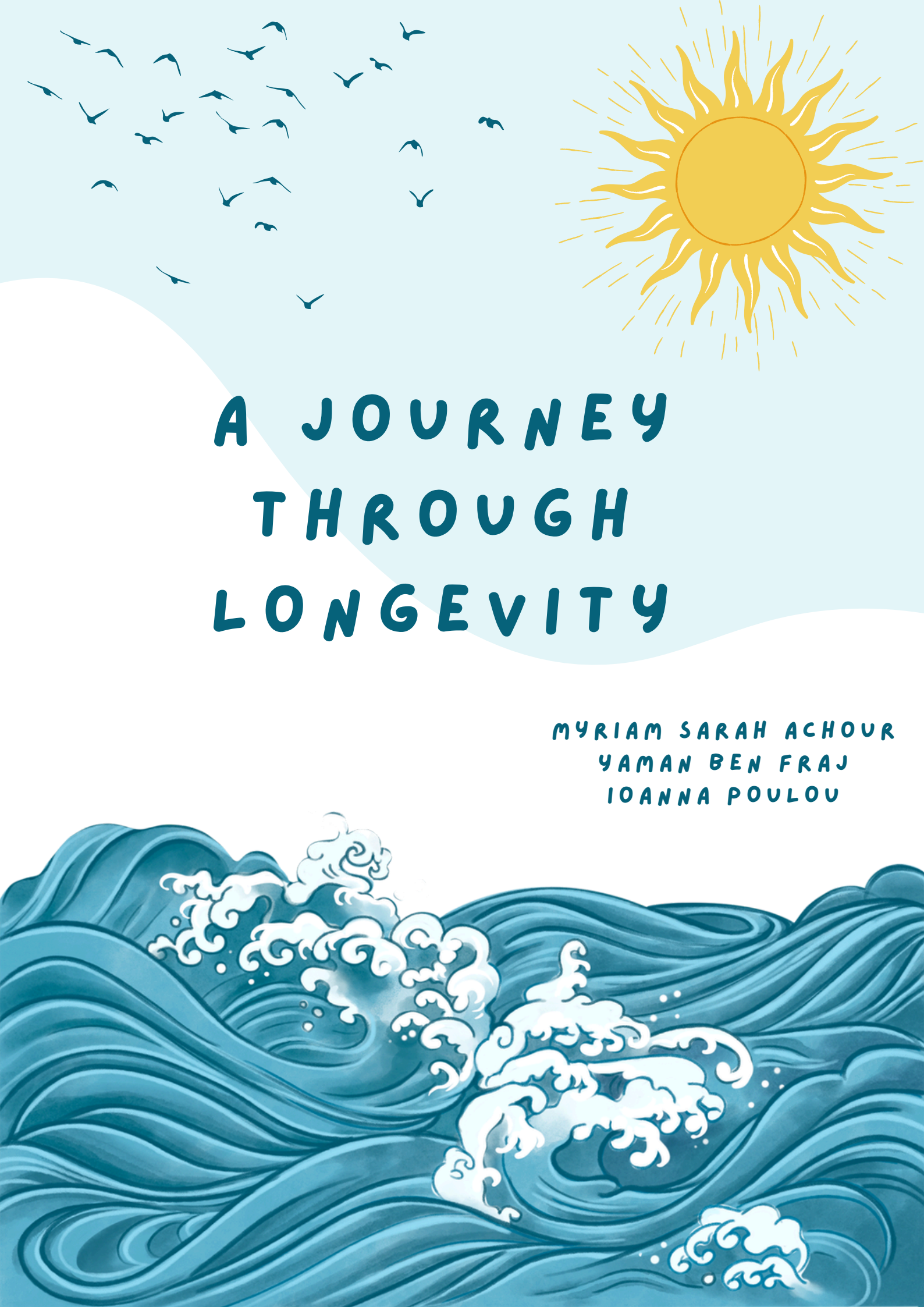




A JOURNEY THROUGH LONGEVITY

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Introduction

In today's fast-moving world, life often feels like a race with no finish line. People chase money, success and comfort, while spending less time with friends and family and living under constant stress. But although modern life may offer convenience, it does not always offer what matters most: **a longer and healthier life**.

Some places, however, tell a different story. Around the world, there are five remarkable regions known as Blue Zones (Okinawa, Ikaria, Sardinia, Nicoya Peninsula and Loma Linda), where people found the formula for living longer. Their secret is not found in expensive treatments, advanced technology, or strict routines, but in nine simple everyday habits: moving naturally, having a clear sense of purpose, slowing down and reducing stress, eating until they are 80% full, following a plant-based diet, drinking alcohol moderately, belonging to a faith-based community, putting family first and maintaining strong social circles.

Inspired by these extraordinary places, our project explores what the Blue Zones can teach the rest of the world. First, we aim to bring these communities and their habits into the spotlight. Then, through an engaging visualization, we investigate whether similar habits appear at a country level and whether they are connected to higher life expectancy.

At the heart of our project lies one powerful question: **can the secrets of the world's longest-living communities be seen across the globe?** By connecting lifestyle data with longevity data, we hope to explore whether we really need to live in a Blue Zone to benefit from its lifestyle, or whether parts of that lifestyle can be recreated wherever we are.

The Path to the Final Result



DATA



EDA



DESIGN



IMPLEMENTATION

1. DATA

Our journey began with extensive research to identify suitable and reliable datasets. One of the main challenges we faced was finding ways to translate the Blue Zone habits into numerical values that could be used in our analysis. After a lot of research, we realized that not all nine habits could be measured clearly or consistently across countries.

For this reason, we decided to focus only on the habits that could be represented with available data (natural movement, stress reduction, the “80% Rule”, plant-based diet and wine consumption). To make this analysis possible, we used the following datasets that best matched these habits:

- **Wine Consumption:** Wine consumption per capita data from Our World in Data to represent wine consumption.
- **Life Expectancy:** Country-level life expectancy data from Our World in Data as the main indicator of longevity.
- **Activity:** Activity Inequality dataset to represent the natural movement level across countries.
- **Happiness:** World Happiness Report to represent stress reduction.
- **Diet Dataset:** Calories intake by food commodity types per capita per day in a given country from Our World in Data from which we derived the:
 - **Plant-Based Diet Ratio:** Sum of daily calories from plant-based foods divided by the total daily calorie intake.
 - **80% Rule:** Optimal intake (≤ 2100 kcal) scores $\text{intake}/2100$. Dangerously low (< 1200 kcal) get a 0 and excessive intakes are penalized, using the formula:
 - $\text{Score} = \max(0, 1 - ((\text{Calories} - 2100)/2600))$.

2. EDA

After finalizing our idea and selecting the datasets, we conducted an Exploratory Data Analysis to better understand the data. We first checked whether the datasets were clean, consistent and free of missing values. Then, we created several visualizations to explore the data, identify patterns, observe trends and detect possible outliers. This step helped us understand the structure of the data before moving on to deeper analysis and visualization.

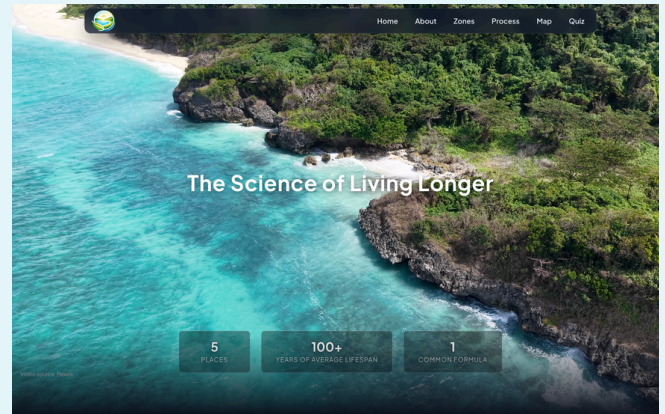
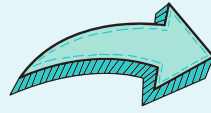
3. DESIGN AND IMPLEMENTATION

Our website and visualizations were built entirely from scratch using HTML, CSS, JavaScript, D3.js and TopoJSON.

The final website includes six main components, each based on the design sketches and structure we developed for Milestone 2.

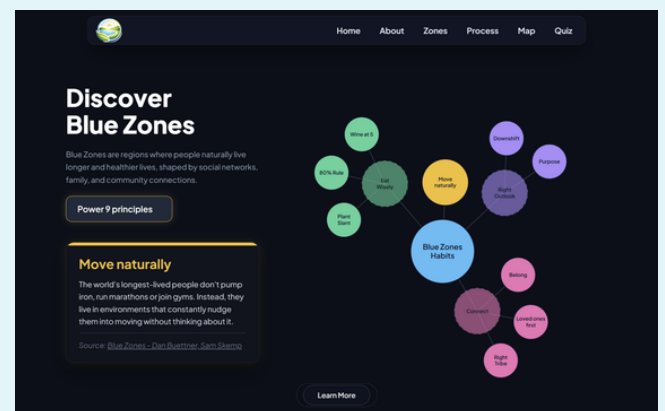
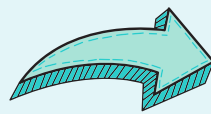
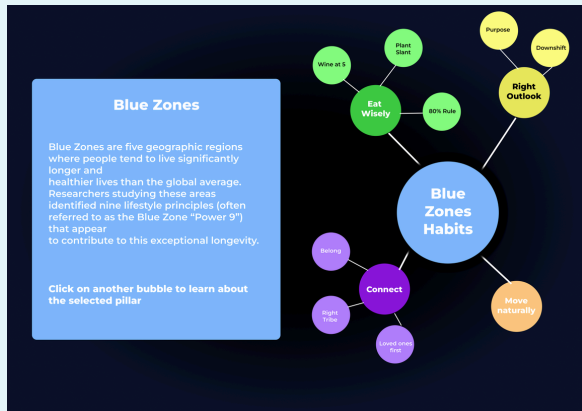
We recommend using Google Chrome when visiting our website

First Glimpse into the Journey



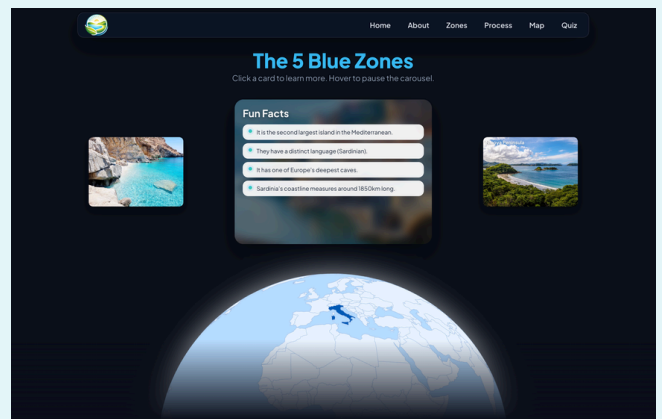
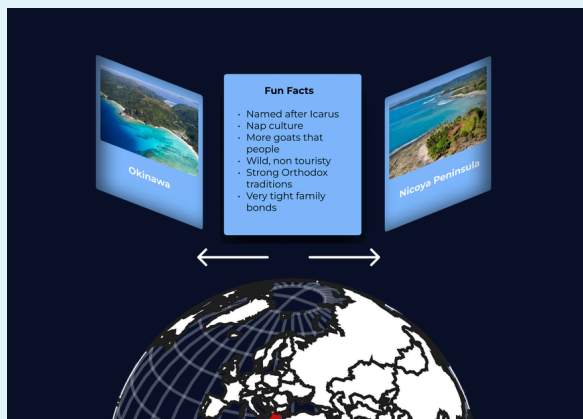
The first thing users see when they enter our website is a video of Okinawa's coast. Although the video has no sound, it still helps capture the user's attention and creates a calm first impression. We chose to begin this way because it immediately introduces the atmosphere and setting of the project, making the website feel more immersive before users explore the rest of the content. It also encourages a sense of calmness, reflecting the Blue Zone idea of slowing down and not rushing through life. Compared to our initial sketch, we also added three introductory boxes: "5 Places" "100+ Years of Average Lifespan" and "1 Common Formula" which give users a quick overview before they continue exploring the website.

The Habits Behind Longevity



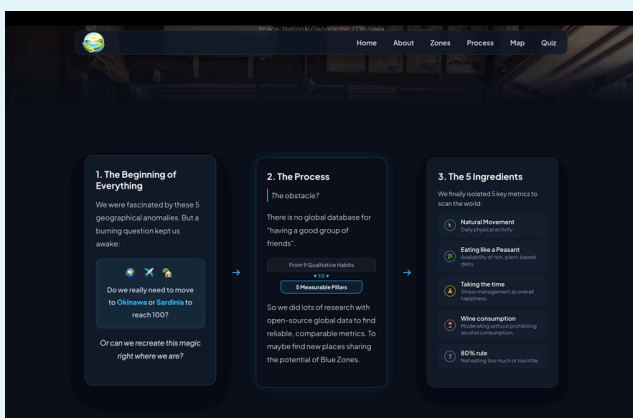
When discussing our project idea with friends, we realized that the concept of "Blue Zones" is not widely known, despite how fascinating it is. Therefore, it became essential for us to not only design an engaging website but also to educate users on this topic. To achieve this, we implemented an interactive force-directed chart. This animated visualization allows users to instantly identify the nine pillars of Blue Zones and their main categories. Clicking on any bubble dynamically updates the information box on the left with a clear description of that pillar. This content is an enhanced and refined version of the information available on the [Blue Zones website](#).

A Tour of the Blue Zones



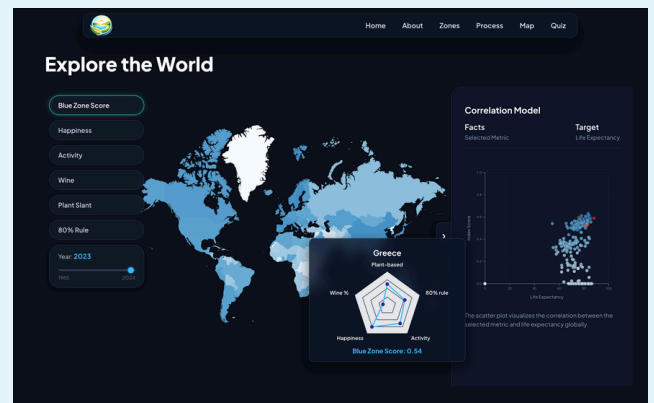
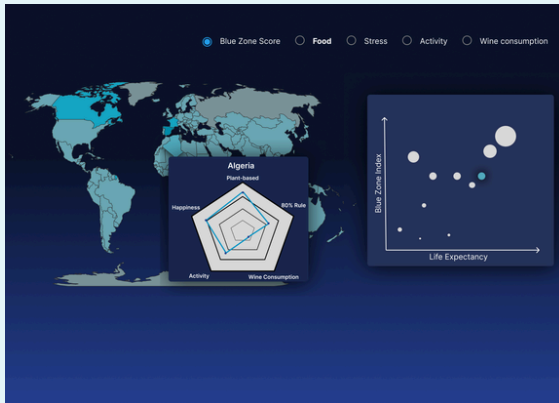
After introducing users to Blue Zones and the pillars of longevity, a natural question emerged: **now that they understand what Blue Zones are, do they know where these places are located?** To answer this, we wanted to create more than just a map. For this reason, we designed an auto-rotating globe that highlights the country in which each Blue Zone belongs to, helping users connect every region with its place in the world. To make the experience richer and more visual, we paired the globe with interactive cards that display an image of each Blue Zone, giving users a clearer sense of what these places actually look like. By clicking on a card, users can reveal more information about the region, transforming the experience from a simple geographic overview into an engaging journey through the cultures of these unique places.

Shaping the Question



As mentioned earlier, not all nine Blue Zone pillars could be included in our final analysis. For this reason, we added this section as a bridge between the storytelling part of the website and the main visualization. Its purpose is to present users with the key question we wanted to answer, while also explaining why we focused only on these specific pillars. This helps users understand the logic behind our choices before they move on to exploring the rest of the website.

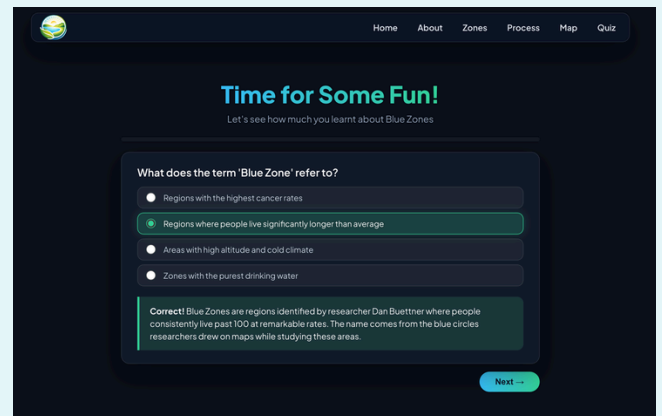
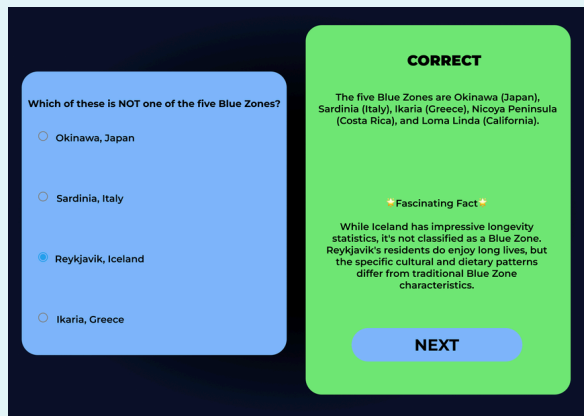
Which Country Lives Like a Blue Zone?



This is the main visualization of our project and the point where our central question truly comes to life: **can Blue Zone lifestyle patterns be seen across the globe and are they connected to longer life?** Through an interactive world map, users can explore how closely each country aligns with the habits found in Blue Zones, with a color gradient showing low to high alignment. Using the buttons, they can focus on one pillar at a time or switch to the combined Blue Zone Score, where all five pillars are weighted equally and contribute to one overall index. When a country is selected, a radar chart reveals its performance across the five pillars, helping users quickly understand whether it follows a balanced Blue Zone-like lifestyle. Based on our global index and latest data, **Sweden shows in 2024 the strongest connection with centenarian-like lifestyle patterns**, making it the country that comes closest to a Blue Zone profile in our analysis.

To take the exploration further, we added a scatter plot that compares each country's selected score with its life expectancy. When users click on a country on the map, the corresponding dot in the scatter plot is highlighted, which makes it easier to locate that country in the chart. Unlike our first sketches, this chart is hidden by default and can be opened with an expand button. This keeps the interface clean and allows users to focus on the map before moving into deeper analysis. We also added a time slider, enabling users to explore how both lifestyle patterns and life expectancy change over time. The time slider's available interval dynamically adjusts based on the selected pillar. We implemented this restriction because the historical depth of our datasets varies, ensuring users only explore years with valid data and preventing empty map states.

A Final Longevity Challenge



As users reach the end of the website, we wanted to close the experience in a light, playful and memorable way. That is why we created a short quiz, giving users the chance to have fun while reflecting on what they have learned throughout their longevity journey. To make the quiz more than just a test, every answer, whether correct or incorrect, is followed by a brief explanation that reinforces the learning behind each question. We intentionally chose simple questions because the goal of the website is not to challenge users with difficult details, but to help them remember the most important ideas about Blue Zones. With five questions, matching the five Blue Zones, the quiz becomes a final moment of interaction that leaves users entertained and with knowledge they can carry forever with them.

Challenges

The most important challenge we faced was standardizing country names across the different datasets. Each dataset used slightly different naming conventions for the same countries, such as “USA” and “United States”. In addition, the country names used in the website’s map JSON file did not always match the names in our datasets (e.g “United States of America”), which made it difficult to merge everything correctly.

To solve this, we decided to use numeric country codes as a common identifier, since these codes were already included in the map JSON file. However, most of our datasets only contained country names and in some cases ISO codes, but not numeric codes. To bridge this gap, we used an LLM to help match the countries from our datasets with the corresponding numeric codes from the map JSON file and add a new numeric code column to each dataset. Once this was completed, we were able to merge all datasets using the numeric code column, making it much easier to connect the data with both the world map and the scatter plot.

Peer Assessment

The entire project was a **collaborative effort**, with all team members working closely together throughout the process. We regularly reviewed each other's work, shared feedback and supported one another in solving challenges, ensuring that the final result reflected a team effort. Following this, we provide a more detailed analysis of how the work was divided among the team:

Myriam had the initial project idea, led the design and implemented the force-directed chart, the bridge sections and the interactive, dynamic world map with pillars and year slider.

Yaman implemented the cards' carousel and the scatter plot, standardized and imported the data and created the screencast.

Ioanna led the Milestone 2 report and the process book and implemented the homepage, quiz, interactive cards and rotating globe.

All team members contributed to the exploratory data analysis and data cleaning.

Conclusion

In the end, our project is more than a website about living longer. **It is a journey through the places, habits and ideas that show how longevity can become part of everyday life.** By combining storytelling, interactive visualizations, data exploration and playful learning, we aimed to make the concept of Blue Zones both understandable and memorable. From discovering the nine pillars of longevity, to exploring where Blue Zones are located, to allowing users to compare which countries around the world follow similar habits and whether people in those countries actually live longer, each section was designed to guide users step by step through the experience. Finally, the quiz brings the journey to a light close, allowing users to revisit what they learned in a fun way. **Our goal was not only to present information, but to inspire users to think about how small daily habits, strong communities and balanced lifestyles can shape a longer and healthier life.**

Acknowledgment

We acknowledge that we used LLMs to assist with parts of the code and with dataset standardization. For some parts of the code, we also were inspired by / used code from publicly available online examples, such as scatter plot and carousel implementations.